A8.1 QoS signaling

This section describes the additional signaling in the AXI4 protocol to support *Quality of Service* (QoS).

A8.1.1 QoS interface signals

The AXI4 signal set is extended to support two 4-bit QoS identifiers:

AWQOS A 4-bit QoS identifier, sent on the write address channel for each write transaction.

ARQOS A 4-bit QoS identifier, sent on the read address channel for each read transaction.

In this specification, **AxQOS** indicates **AWQOS** or **ARQOS**.

The protocol does not specify the exact use of the QoS identifier. This specification recommends that **AxQOS** is used as a priority indicator for the associated write or read transaction. A higher value indicates a higher priority transaction.

A default value of `0b0000` indicates that the interface is not participating in any QoS scheme.

Note

Additional interpretations of the QoS identifier can be used.

A8.1.2 Master considerations

A master can produce its own **AxQOS** values, and if it can produce multiple streams of traffic it can choose different QoS values for the different streams.

Support for QoS requires a system-level understanding of the QoS scheme in use, and collaboration between all participating components. For this reason, this specification recommends that a master component includes some programmability that can be used to control the exact QoS values used for any given scenario.

If a master component does not support a programmable QoS scheme it can use QoS values that represent the relative priorities of the transactions it generates. These values can then be mapped to alternative system level QoS values if appropriate.

A master that can not produce its own **AxQOS** values must use the default value.

Note

This specification expects that many interconnect component implementations will support programmable registers that can be used to assign QoS values to connected masters. These values replace the QoS values, either programmed or default, supplied by the masters.

A8.1.3 System considerations

QoS signaling as defined in AXI4 can be used with any compatible system-level QoS methodology.

The default system-level implementation of QoS is that any component with a choice of more than one transaction to process selects the transaction with the higher QoS value to process first. This selection only occurs when there is no other AXI constraint that requires the transactions to be processed in a particular order.

Note

This means that the AXI ordering rules take precedence over ordering for QoS purposes.

More sophisticated QoS schemes that are compatible with this default scheme can be implemented.

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